CHAEEUN KIM

E-mail: codms0914@snu.ac.kr | Website: https://dimchae-eun.github.io
Researcher | Seoul National University | Republic of Korea

Research Interests | B*iosensors; Bioelectronics; Wearable sensing; Continuous biomarker monitoring; Point-of-care diagnostics*

Education

Seoul National University, Seoul, Republic of Korea Feb 2026

- M.S. in Applied Bioengineering. Advisor: Prof. Kangwon Lee.
- Thesis: Calibration-Free Potassium Quantification via a Stoichiometric Enzyme Cascade on Laser-Induced Graphene **(Outstanding Thesis Award)**

Chung-Ang University, Seoul, Republic of Korea Feb 2024

- B.S. in Biomedical Engineering.
- Thesis: Trends in Antigen-Antibody-Based Biosensor Technology for Pathogen Diagnosis.

Publications

1. **Chaeeun Kim**, Chaehyung Kim, Yujoon Kim, and Kangwon Lee[*]. Calibration-free potassium quantification via a stoichiometric enzyme cascade on laser-induced graphene. *Biosensors and Bioelectronics* 304, 118572 (2026). DOI: 10.1016/j.bios.2026.118572
2. Cheol Ho Park[†], Yujoon Kim[†], **Chaeeun Kim**[†], Chaehyung Kim[†], Sowon Kim, Kangwon Lee, and Tae-Hyun Yoo[*]. Clinical evaluation of a disposable test strip for potassium level measurement in patients treated with maintenance hemodialysis. *Clinical Journal of the American Society of Nephrology* 21(3), 425–436 (2026). DOI: 10.2215/CJN.0000000934. [†]Equal contribution.
3. Jaeyeon Lee, **Chaeeun Kim**, Chaehyung Kim, Yujoon Kim, Seojin Lee, Kangwon Lee[*], and In Young Kim[*]. Supercapacitive carbon-coated δ-MnO_2 nanosheets as a solid-contact material for self-testing potassium-ion biosensors. *Small* 22(26), e13386 (2026). DOI: 10.1002/smll.202513386

Patent

1. **Strips for measuring glucose and potassium.** Yujoon Kim, **Chaeeun Kim**, Min Hee Park, Jaeho Ham, and Kangwon Lee. Korean Patent No. 10-2757781 (granted Jan 2025).

Research Experience

Advanced Functional Biomaterials Laboratory, Seoul National University Jul 2023 – Present
Advisor: Prof. Kangwon Lee

Researcher Mar 2026 – Present

- Lead translational R&D of a portable potassium-ion reader and disposable test-strip platform in an industry–academia project with THEDONEE Inc.; coordinate sensor validation, measurement-circuit and firmware development, housing design, and external testing. Completed a working prototype, exhibited at BIX Korea 2026 (Oct 2026).
- Fabricate electrospun components for a MOTIE-funded 3D cell-based regenerative therapy project, supporting the development of materials and devices for cell preservation and thawing.

Graduate Researcher Mar 2024 – Feb 2026

- **Calibration-free enzymatic potassium sensor.** Developed an amperometric potassium sensor using a pyruvate kinase/pyruvate oxidase cascade on laser-induced graphene (LIG); led experimental design, optimization, and manuscript preparation. Achieved a 2–8 mM linear range and inter-sensor RSD of 2.41–5.20%; validated in potassium-spiked human serum
- **Clinical evaluation of potassium test strips.** Evaluated disposable potentiometric potassium strips with capillary blood from 40 patients receiving maintenance hemodialysis, in collaboration with Severance Hospital: optimized strips, coordinated the study protocol and IRB, tested patient blood samples, and

performed Bland–Altman and Passing–Bablok analyses in R. Capillary-blood strip readings agreed with central-laboratory serum values (Passing–Bablok slope 1.04; 95% CI, 0.97–1.12).

- **Solid-contact materials.** Integrated carbon-coated δ-MnO_2 nanosheets developed at Ewha Womans University into potassium-sensing strips; evaluated electrochemical performance and batch reproducibility, contributing to a 1.4-fold faster response time than activated-carbon solid contacts over a 10^{-4}–10^{-1} M K^+ range.
- **Storage-stable reference electrode.** Developed a graphene nanoplatelet/polyvinyl butyral (GNP/PVB)-modified Ag/AgCl reference electrode for screen-printed potassium ion-selective strips; sensitivity remained near-Nernstian after 60 days of storage (55.28 ± 0.70 to 56.53 ± 1.28 mV/decade, n = 5).
- **Dual glucose–potassium test strip** (Seoul R&D Program, Seoul Business Agency BT230224; Sep 2023 – Aug 2025). Responsible for potassium-sensor design, experiments, data analysis, and integration into the dual-analyte strip; resulted in a granted Korean patent.

Undergraduate Research Intern Jul 2023 – Feb 2024

- Fabricated, characterized, and optimized potassium ion-selective electrodes.

Translational Multiscale Biofluidic Laboratory, UNIST Jul 2022

Undergraduate Research Intern (U-SURF program); Advisor: Prof. Joo Hun Kang

- Trained in magnetic blood cleansing, μFISH-based bacterial identification, and vascularized microfluidic networks.

Department of Preventive Medicine, Seoul National University College of Medicine Jun 2021 – Aug 2021

Undergraduate Research Intern; Advisor: Prof. Aesun Shin

- Reviewed literature on genetic variants associated with colorectal cancer susceptibility in Asian populations.

Conference Presentations

1. **Chaeeun Kim**, Kangwon Lee*. "Long-Term Storage-Stable Potassium Sensing Enabled by a PVB-Enhanced Reference Electrode." SNU BioDay, Sep 2025. Poster.
2. **Chaeeun Kim**, Chaehyung Kim, Yujoon Kim, Kangwon Lee*. "Polyvinyl Butyral (PVB)-Enhanced Reference Electrode for Stable Potassium Sensing in Point-of-Care (POC) Applications." NANO KOREA, Jul 2025. Poster.
3. **Chaeeun Kim**, Chaehyung Kim, Yujoon Kim, Kangwon Lee*. "Polyvinyl Butyral (PVB)-Enhanced Reference Electrode for Stable Potassium Sensing in Point-of-Care (POC) Applications." Korean BioChip Society, Nov 2024. Poster **(Best Poster Presentation Award)**.

Honors and Scholarships

Outstanding Thesis Award, *Graduate School of Convergence Science and Technology, SNU* Feb 2026
Lecture & Research Scholarship, *Seoul National University* Fall 2025
Best Poster Presentation Award, *Korean BioChip Society* Nov 2024
Merit-based Scholarship, *Seoul National University* Fall 2024
Audience Choice Award, *ICC CORE-ATHON Startup Competition* Dec 2022
Scholarship, *Ulsan Institute for Talent and Lifelong Education* Fall 2021

Research & Technical Skills

Electrochemical Methods: Chronoamperometry (CA), Cyclic Voltammetry (CV), Potentiometry

Fabrication & Assay Development: Laser-Induced Graphene (LIG) Fabrication, Ion-Selective Electrode (ISE) Fabrication, Enzyme-Cascade Optimization, O_2 Plasma Treatment, Electrospinning, Microplate-Based Absorbance Assays (UV-Vis)

Materials Characterization: Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), Raman Spectroscopy

Analytical & Clinical Validation: Selectivity and Reproducibility Testing, Clinical Sample Analysis, Method Comparison (Bland–Altman, Passing–Bablok)

Software: R, OriginPro, GraphPad Prism, ImageJ, AutoCAD